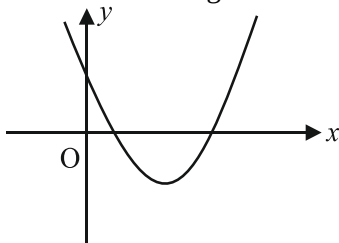
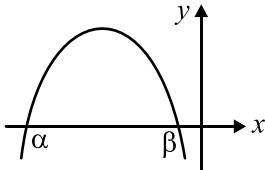
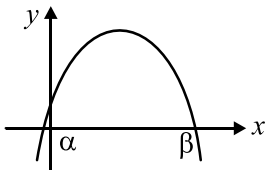




Prayas JEE 2.0 (2024)

Quadratic Equations

DPP-04

- Let a, b and c be real numbers such that $4a + 2b + c = 0$ and $ab > 0$. Then the equation $ax^2 + bx + c = 0$ has
 - Real roots
 - Imaginary roots
 - Exactly one root
 - None of these
- The values of k , so that the equation $2x^2 + kx - 5 = 0$ and $x^2 - 3x - 4 = 0$ may have one root in common.
 - $-3, -1$
 - $-3, \frac{-27}{4}$
 - $-1, -2$
 - $3, \frac{27}{4}$
- The graph of $y = ax^2 + bx + c$ is shown in figure. Which of the following does **NOT** hold good?
 
 - $ab^2c^3 > 0$
 - $ab^3c^2 < 0$
 - $ab^3c^5 > 0$
 - $b^2 > 4ac$
- For following graphs of $y = ax^2 + bx + c$ with $a, b, c \in R$, comment on the sign of :
 - a
 - b
 - c
 - D
 - $\alpha + \beta$
 - $\alpha \beta$
 - 
 - 
- Draw graph of $f(x) = 1 - (x^2 + 4x)$.
- If $f(x)$ is non-negative monic quadratic function which touches x -axis at $x = -2$ then find $f(5) =$
- If $y = -2x^2 - 6x + 9$, then
 - maximum value of y is -11 and it occurs at $x = 2$
 - minimum value of y is -11 and it occurs at $x = 2$
 - maximum value of y is 13.5 and it occurs at $x = -1.5$
 - minimum value of y is 13.5 and it occurs at $x = -1.5$
- If the real and distinct roots α and β of the equation $x^2 + 1 = \frac{x}{a}$ satisfy $|\alpha^2 - \beta^2| > \frac{1}{a}$, then ' a ' belongs to
 - $\left(-\frac{1}{2}, 0\right) \cup \left(0, \frac{1}{2}\right)$
 - $\left(-\frac{1}{2}, 0\right) \cup \left(0, \frac{1}{\sqrt{5}}\right)$
 - $\left(-\frac{1}{\sqrt{5}}, 0\right) \cup \left(0, \frac{1}{\sqrt{5}}\right)$
 - None of these
- If α, β are the roots of the equation $x^2 - 3x + 5 = 0$ and γ, δ are the roots of the equation $x^2 + 5x - 3 = 0$, then the equation whose roots are $\alpha\gamma + \beta\delta$ and $\alpha\delta + \beta\gamma$, is
 - $x^2 - 15x - 158 = 0$
 - $x^2 + 15x - 158 = 0$
 - $x^2 - 15x + 158 = 0$
 - $x^2 + 15x + 158 = 0$
- Let α, β be the roots of the equation $x^2 - x + p = 0$ and γ, δ be the roots of the equation $x^2 - 4x + q = 0$. If α, β, γ and δ are in GP, the integral values of p and q respectively, are **[Challenger]**
 - $-2, -32$
 - $-2, 3$
 - $-6, 3$
 - $-6, -32$



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

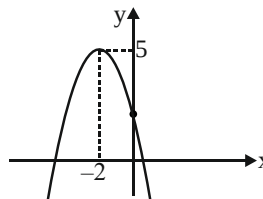
Answer Key

1. (1)
2. (2)
3. (3)
4. (1) (i) $a < 0$,
 (ii) $b < 0$,
 (iii) $c < 0$,
 (iv) $D > 0$;
 (v) $\alpha + \beta < 0$,
 (vi) $\alpha\beta > 0$
 (2) $a < 0$,
 (ii) $b > 0$,
 (iii) $c > 0$,
 (iv) $D > 0$;

(v) $\alpha + \beta > 0$,

(vi) $\alpha\beta < 0$

5.



6. (49)

7. (3)

8. (2)

9. (4)

10. (1)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>